Tutorial 7 - Solutions

1. (a) Let μ_1 and μ_2 denote the mean reaction times of the populations represented by the group using a mobile phone and the control group respectively. It is convenient to set the hypothesis that we want to test as our alternative hypothesis, i.e. test

$$H_0 : \mu_1 = \mu_2 \quad v \quad H_1 : \mu_1 > \mu_2$$

as this will allow us to fully determine the sampling distribution of the test statistic under H_0. The two samples are large enough to use the CLT, so we have that

$$\frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{633.5 - 571.2}{\sqrt{\frac{98.1^2}{31} + \frac{89.8^2}{31}}} = 2.608$$

should be a value from $N(0,1)$ under H_0. Using tables we find that the critical region is given by the values $z > z_{0.01} = 2.326$ (since this is a one-sided test), and therefore we *reject* H_0 at the 1% level of significance in favour of the alternative hypothesis that the mean reaction time for drivers that use a phone is greater than for those who don't.

(b) We have

$$p\text{-value} = \mathbb{P}(Z > 2.608) = 0.005 \quad \text{where } Z \sim N(0,1).$$

This gives the probability to observe a value as or more extreme than 2.608 under H_0, and since p-value < 0.01 we have *strong* evidence against H_0 and the conclusion is the same as in (a).

2. The variances of the two samples are

$$s_x^2 = \frac{1}{8}(4000 - 180^2/9) = 50 \quad \text{and} \quad s_y^2 = \frac{1}{9}(4930 - 220^2/10) = 10.$$

Under the null hypothesis the value

$$F = \frac{s_x^2}{s_y^2} = 5$$

should come from an $F_{8,9}$ distribution. With a one-sided test, we have that the critical region is given by $F > F_{8,9,0.05} = 3.23$. Since the observed value of the statistic is greater than 3.23 we reject the null hypothesis and conclude that $\sigma_x^2 > \sigma_y^2$.

3. (a) We want to test $H_0 : \mu_B - \mu_A = 0 \quad v \quad H_1 : \mu_B - \mu_A \neq 0$ at $\alpha = 0.05$. Assuming normality and equality of variances, and using the sample values computed in Tutorial 6 we have:

$$t = \frac{\bar{x}_B - \bar{x}_A}{\sqrt{s_p^2(1/n_1 + 1/n_2)}} = \frac{8.444 - 8.056}{\sqrt{1.294 \times 2/9}} = 0.7236.$$

The *critical values* (i.e. the values determining the critical region) are $t_{16,0.025} = 2.12$ and -2.12, and therefore we *do not reject* the hypothesis that $\mu_B = \mu_A$ at the 5% level of significance.

(Notice also that p-value $= 2\mathbb{P}(t_{16} > 0.7236) = 0.48$. This can be computed in `R` as `2*(1-pt(0.7236,16))`.)

(b) With the data in matched pairs, and assuming again normality of the differences $d_i = x_{B,i} - x_{A,i}$, we have

$$t = \frac{\bar{d}}{\sqrt{s_D^2/n}} = \frac{0.389}{\sqrt{0.111/9}} = 3.503.$$

The critical values are now $t_{8,0.025} = 2.306$ and -2.306, so we now reject the hypothesis that $\mu_B = \mu_A$ at the 5% level of significance!

(again notice that p-value $= 2\mathbb{P}(t_8 > 3.503) = 0.008$.)

(c) The 95% CI in part (a) of Q4, Tutorial 5 includes the value 0, whereas in part (b) (when data are considered paired) it doesn't. This agrees with our conclusions in parts (a) and (b) here.

4. If X is the number of policies giving claims we have $X \sim \text{Bin}(150, p)$ and we want to test $H_0 : p = 0.5 \quad v \quad H_1 : p > 0.5$. Under the null hypothesis $X \sim \text{Bin}(150, 0.5)$ and using the CLT $X \sim N(75, 37.5)$ approximately. Then

$$\begin{aligned} p\text{-value} &= \mathbb{P}(X \geq 83) = \mathbb{P}(Z > \frac{83 - 75}{\sqrt{37.5}}) \quad \text{where } Z \sim N(0,1) \\ &= \mathbb{P}(Z > 1.306) = 0.096. \end{aligned}$$

Since p-value > 0.01 we do not reject the hypothesis that $p = 0.5$.

Note that you can also use a *continuity correction*, that will give

$$p\text{-value} = \mathbb{P}(X > 82.5) = \mathbb{P}(Z > \frac{82.5 - 75}{\sqrt{37.5}}) = \mathbb{P}(Z > 1.2247) = 0.110$$

which is the same (to 3 d.p.) as the exact answer obtained with the binomial distribution in `R` as `1-pbinom(82,150,0.5) = 0.110`.

5. We assume that the two samples are independent and come from two $\text{Bin}(n_1, p_1)$ and $\text{Bin}(n_2, p_2)$ populations. We want to test $H_0 : p_1 = p_2 \quad v \quad H_1 : p_1 \neq p_2$ and use the statistic

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$$

with $\hat{p}$ being the common estimate

$$\hat{p} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1 + n_2} = \frac{3000 \times 0.077 + 3120 \times 0.059}{6120} = 0.0678,$$

giving

$$z = \frac{0.077 - 0.059}{\sqrt{0.0678(1-0.0678)(\frac{1}{3000} + \frac{1}{3120})}} = 2.81.$$

The critical values are $z_{0.025} = 1.96$ and -1.96, so we reject the null hypothesis at $\alpha = 0.05$, i.e. we have evidence that there has been a change in the preferences of consumers.

Exam Style Question

One way to measure a person's fitness is to measure their body fat percentage. Data are measured for a random selection of people who did two different workout plans at a gym three times a week for a three-month period; at the end of which their trainer measured the body fat. The table below shows the data.

Plan A	8.5	20.8	23.1	22.6	24.1	15.5	15.4	13.7
Plan B	24.6	6.8	10.4	10.3	16.4	8.3	13.2	17.7

(a) We perform the test:

$$H_0 : \mu_A = \mu_B \text{ versus } H_1 : \mu_A \neq \mu_B.$$

The test statistic is

$$T(\underline{X}_A, \underline{X}_B) = \frac{\bar{X}_A - \bar{X}_B}{S_p\sqrt{\frac{1}{n_A} + \frac{1}{n_B}}}$$

which is distributed according to $t_{n_A+n_B-2}$ under the null hypothesis.

Note that

$$\bar{x}_A = 17.96, \quad \bar{x}_B = 13.46.$$

Our observed value of the test statistic is

$$T(\underline{x}_A, \underline{x}_B) = \frac{17.96 - 13.46}{\sqrt{32.51}\sqrt{\frac{1}{8} + \frac{1}{8}}} = 1.58.$$

The p-value is the probability of seeing a value of $T(\underline{X}_A, \underline{X}_B)$ that is as or more extreme then $T(\underline{x}_A, \underline{x}_B)$ assuming H_0 is true, i.e.

$$\mathbb{P}(|T(\underline{X}_A, \underline{X}_B)| \geq 1.58; \mu_a = \mu_B) = 2\mathbb{P}(T14 \geq 1.8)$$

where $T_{14} \sim t_{14}$. Using the statistical tables we obtain the p-value

$$p = 2(1 - 0.934) = 0.132.$$

This means that there is insufficient evidence to reject H_0.

[5 marks]

(b) We perform the test:

$$H_0 : \sigma_A = \sigma_B \text{ versus } H_1 : \sigma_A \neq \sigma_B.$$

We use the test statistic:

$$T(\underline{X}_A, \underline{X}_B) = \frac{S_A^2}{S_B^2}$$

which is F_{n_A-1,n_B-1} distributed if the null hypothesis is true. We reject the null hypothesis if we observe a value of the test statistic outside the range $(F_{n_A-1,n_B-1,0.975}, F_{n_A-1,n_B-1,0.025})$. From the statistical tables

$$F_{7,7,0.025} = 4.995$$
$$F_{7,7,0.975} = \frac{1}{F_{7,7,0.025}} = 0.2002$$

Our observed value of the test statistic is 0.89 and as this does not fall in the rejection region we have insufficient evidence to reject the null hypothesis.

(3 marks)

(c) What are the main assumptions in assessing whether or not the test of part (a) is valid? Give reasons whether or not these assumptions hold for the above data.

There are two main assumptions for (a) to be valid:

- The population variances are equal;
- The populations are normally distributed.

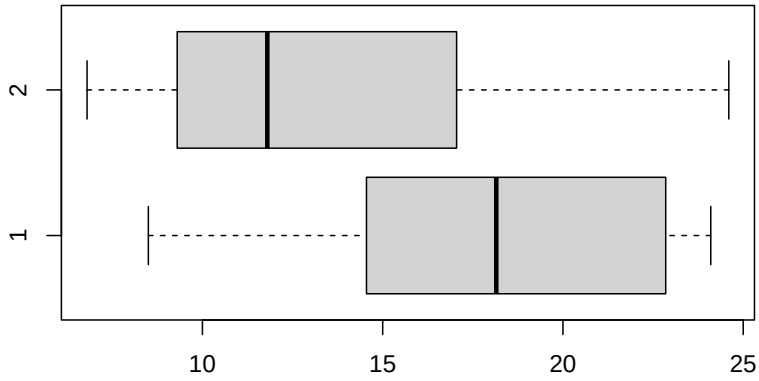
While it is not possible to verify these assumptions as we do not have access to the population we can check to see if there is any reason to doubt the assumptions based on our sample.

We showed in (b) that there is insufficient evidence to conclude that the variances are not equal.

We can use box plots to see if the normality assumption is reasonable.

[2 marks]

[10 marks]


2
1
10
15
20
25